# OPi_CM5_CORE JP1 100PIN

| Description | Pin Name | Pin Net | PIN NO. | | Signal | Pin Name | Description |
|---|---|---|---|---|---|---|---|
| **Ground (0V)** | | **GND** | **1** | **2** | **GND** | | **Ground (0V)** |
| Floatimg | | Reserved | 3 | 4 | Reserved | | Floatimg |
| Floatimg | | Reserved | 5 | 6 | Reserved | | Floatimg |
| **Ground (0V)** | | **GND** | **7** | **8** | **GND** | | **Ground (0V)** |
| Floatimg | | Reserved | 9 | 10 | Reserved | | Floatimg |
| Floatimg | | Reserved | 11 | 12 | Reserved | | Floatimg |
| **Ground (0V)** | | **GND** | **13** | **14** | **GND** | | **Ground (0V)** |
| Floatimg | | Reserved | 15 | 16 | Reserved | | Floatimg |
| Floatimg | | Reserved | 17 | 18 | Reserved | | Floatimg |
| Floatimg | | Reserved | 19 | 20 | Reserved | | Floatimg |
| GPIO Typically a 3.3V | CIF_D4/BT1120_D4/DDRPHY_CH1_DTB_0/I2C3_SCL_M2/UART0_RX_M2/SPI2_MISO_M1/GPIO4_A4_d | GPIO4_A4 | 21 | 22 | **GND** | | **Ground (0V)** |
| **Ground (0V)** | | **GND** | 23 | 24 | GPIO1_C0 | I2C3_SDA_M0/UART3_RX_M0/SPI4_MISO_M0/GPIO1_C0_z | GPIO Typically a 3.3V，3.3V or 1.8V |
| GPIO Typically a 3.3V，3.3V or 1.8V | PDM1_SDI1_M1/SPI2_CS1_M0/GPIO1_B0_u | GPIO1_B0 | 25 | 26 | GPIO1_C1 | I2C3_SCL_M0/UART3_TX_M0/SPI4_MOSI_M0/GPIO1_C1_z | GPIO Typically a 3.3V，3.3V or 1.8V |
| GPIO Typically a 3.3V，3.3V or 1.8V | PDM1_SDI2_M1/SPI0_MISO_M2/GPIO1_B1_d | GPIO1_B1 | 27 | 28 | GPIO1_A2 | VOP_POST_EMPTY/I2C4_SDA_M3/UART6_RTSN_M1/PWM0_M2/SPI4_CLK_M2/GPIO1_A2_d | GPIO Typically a 3.3V，3.3V or 1.8V |
| GPIO Typically a 3.3V，3.3V or 1.8V | PDM1_SDI3_M1/UART4_RX_M2/SPI0_MOSI_M2/GPIO1_B2_d | GPIO1_B2 | 29 | 30 | GPIO1_A3 | I2C4_SCL_M3/UART6_CTSN_M1/PWM1_M2/SPI4_CS0_M2/GPIO1_A3_d | GPIO Typically a 3.3V，3.3V or 1.8V |
| GPIO Typically a 3.3V，3.3V or 1.8V | PDM1_CLK1_M1/SATA0_ACT_LED_M1/UART4_TX_M2/SPI0_CLK_M2/GPIO1_B3_d | GPIO1_B3 | 31 | 32 | **GND** | | **Ground (0V)** |
| **Ground (0V)** | | **GND** | 33 | 34 | GPIO1_A4 | SPI2_MISO_M0/GPIO1_A4_d | GPIO Typically a 3.3V，3.3V or 1.8V |
| GPIO Typically a 3.3V | MIPI_CAMERA3_CLK_M0/I2C8_SCL_M2/UART1_RTSN_M1/PWM14_M2/GPIO1_D6_u | GPIO1_D6 | 35 | 36 | GPIO1_D7 | MIPI_CAMERA4_CLK_M0/I2C8_SDA_M2/UART1_CTSN_M1/PWM15_IR_M3/GPIO1_D7_u | GPIO Typically a 3.3V |
| GPIO Typically a 3.3V，3.3V or 1.8V | PDM1_CLK0_M1/UART7_RX_M2/SPI0_CS0_M2/GPIO1_B4_u | GPIO1_B4 | 37 | 38 | GPIO1_A5 | HDMI_TX0_HPD_M0/SPI2_MOSI_M0/GPIO1_A5_d | GPIO Typically a 3.3V，3.3V or 1.8V |
| GPIO Typically a 3.3V，3.3V or 1.8V | UART7_TX_M2/SPI0_CS1_M2/GPIO1_B5_u | GPIO1_B5 | 39 | 40 | GPIO1_A6 | SPI2_CLK_M0/GPIO1_A6_d | GPIO Typically a 3.3V，3.3V or 1.8V |
| GPIO Typically a 3.3V，3.3V or 1.8V | MIPI_CAMERA1_CLK_M0/SPDIF0_TX_M0/I2C5_SCL_M3/UART1_TX_M1/GPIO1_B6_u | GPIO1_B6 | 41 | 42 | **GND** | | **Ground (0V)** |
| **Ground (0V)** | | **GND** | 43 | 44 | GPIO1_A7 | PDM1_SDI0_M1/PCIE20X1_1_PERSTN_M2/PWM3_IR_M3/SPI2_CS0_M0/GPIO1_A7_u | GPIO Typically a 3.3V，3.3V or 1.8V |
| GPIO Typically a 3.3V，3.3V or 1.8V | MIPI_CAMERA2_CLK_M0/SPDIF1_TX_M0/SATA2_ACT_LED_M1/I2C5_SDA_M3/UART1_RX_M1/PWM13_M2/GPIO1_B7_u | GPIO1_B7 | 45 | 46 | GPIO1_C4 | PDM0_CLK1_M0/I2C2_SDA_M3/PWM11_IR_M2/SPI4_CS1_M0/GPIO1_C4_d | GPIO Typically a 3.3V，3.3V or 1.8V |
| GPIO Typically a 3.3V，3.3V or 1.8V | PMIC_SLEEP4/GPIO0_C2_d | GPIO0_C2 | 47 | 48 | GPIO1_C6 | PDM0_CLK0_M0/I2C4_SDA_M4/PWM15_IR_M2/GPIO1_C6_d | GPIO Typically a 3.3V，3.3V or 1.8V |
| GPIO Typically a 3.3V，3.3V or 1.8V | PDM0_SDI0_M0/SPI1_CS1_M2/GPIO1_D5_d | GPIO1_D5 | 49 | 50 | GPIO1_D0 | I2S0_SDO1/I2C7_SCL_M0/UART6_TX_M2/SPI1_MISO_M2/GPIO1_D0_d | GPIO Typically a 3.3V，3.3V or 1.8V |
| GPIO Typically a 3.3V，3.3V or 1.8V | SDMMC_D1/PDM1_SDI2_M0/JTAG_TMS_M1/I2C3_SDA_M4/UART2_RX_M1/PWM9_M1/GPIO4_D1_u | UART2_RX | 51 | 52 | **GND** | | **Ground (0V)** |
| **Ground (0V)** | 0 | **GND** | 53 | 54 | GPIO1_D1 | I2S0_SDO2/I2S0_SDI3/PDM0_SDI1_M0/I2C7_SDA_M0/UART6_RX_M2/SPI1_MOSI_M2/GPIO1_D1_d | GPIO Typically a 3.3V，3.3V or 1.8V |
| GPIO Typically a 3.3V，3.3V or 1.8V | SDMMC_D0/PDM1_SDI3_M0/JTAG_TCK_M1/I2C3_SCL_M4/UART2_TX_M1/PWM8_M1/GPIO4_D0_u | UART2_TX | 55 | 56 | GPIO1_D2 | I2S0_SDO3/I2S0_SDI2/PDM0_SDI2_M0/I2C1_SCL_M4/UART4_TX_M0/PWM0_M1/SPI1_CLK_M2/GPIO1_D2_d | GPIO Typically a 3.3V，3.3V or 1.8V |
| SDCARD CLK signal | SDMMC_CLK/PDM1_CLK0_M0/TEST_CLKOUT_M0/MCU_JTAG_TMS_M0/CAN0_RX_M1/UART5_TX_M0/GPIO4_D5_d | SDMMC_CLK | 57 | 58 | GPIO1_D3 | I2S0_SDI1/PDM0_SDI3_M0/I2C1_SDA_M4/UART4_RX_M0/PWM1_M1/SPI1_CS0_M2/GPIO1_D3_d | GPIO Typically a 3.3V，3.3V or 1.8V |
| **Ground (0V)** | | **GND** | 59 | 60 | **GND** | | **Ground (0V)** |
| SDCARD Data3 signal | SDMMC_D3/PDM1_SDI0_M0/JTAG_TMS_M0/I2C8_SDA_M0/UART5_RTSN_M0/PWM10_M1/GPIO4_D3_u | SDMMC_D3 | 61 | 62 | SDMMC_CMD | SDMMC_CMD/PDM1_CLK1_M0/MCU_JTAG_TCK_M0/CAN0_TX_M1/UART5_RX_M0/PWM7_IR_M1/GPIO4_D4_u | SDCARD CMD signal |
| SDCARD Data0 signal | SDMMC_D0/PDM1_SDI3_M0/JTAG_TCK_M1/I2C3_SCL_M4/UART2_TX_M1/PWM8_M1/GPIO4_D0_u | SDMMC_D0 | 63 | 64 | Reserved | | Do not Connect anything to this pin. |
| **Ground (0V)** | | **GND** | 65 | 66 | **GND** | | **Ground (0V)** |
| SDCARD Data1 signal | SDMMC_D1/PDM1_SDI2_M0/JTAG_TMS_M1/I2C3_SDA_M4/UART2_RX_M1/PWM9_M1/GPIO4_D1_u | SDMMC_D1 | 67 | 68 | SARADC_IN1 | SARADC_IN1 | RECOVERY signal，Internally pulled up via 10K to1.8V |
| SDCARD Data2 signal | SDMMC_D2/PDM1_SDI1_M0/JTAG_TCK_M0/I2C8_SCL_M0/UART5_CTSN_M0/GPIO4_D2_u | SDMMC_D2 | 69 | 70 | Reserved | | Floatimg |
| **Ground (0V)** | | **GND** | 71 | 72 | Reserved | | Floatimg |
| Do not Connect anything to this pin. | | Reserved | 73 | 74 | **GND** | | **Ground (0V)** |
| Output to Power switch for the SDCARD GPIO4_B4 For 3.3V GPIO. | SPI2_CLK_M2/SDMMC_PWREN/PMU_DEBUG/GPIO0_A5_d | SDMMC_PWREN | 75 | 76 | GPIO0_A4 | SDMMC_DET/GPIO0_A4_u | SDMMC0_DET<br>GPIO0_A4,For 1.8V GPIO. |
| **4.75V-5.25V Main power input** | Main power input | **VBUS** | 77 | 78 | **GPIO_VREF** | POWER In,3.3V for 3.3V GPIO,1.8V For 1.8V GPIO | **Power for 3.3V GPIO or for 1.8V GPIO.** |
| **4.75V-5.25V Main power input** | Main power input | **VBUS** | 79 | 80 | I2C6_SCL_M3 | MIPI_CAMERA0_CLK_M0/SPDIF1_TX_M1/I2S1_SDO0_M0/SATA2_ACT_LED_M0/DDRPHY_CH2_DTB1/I2C6_SCL_M3/UART8_RX_M0/SPI0_CS1_M1/GPIO4_B1_u | IIC Clock pin<br>GPIO4_B1, for 3.3V |
| **4.75V-5.25V Main power input** | Main power input | **VBUS** | 81 | 82 | I2C6_SDA_M3 | CIF_CLKIN/BT1120_CLKOUT/I2S1_SDI3_M0/DDRPHY_CH2_DTB0/I2C6_SDA_M3/UART8_TX_M0/SPI2_CS1_M1/GPIO4_B0_d | IIC Data pin<br>GPIO4_B0, for 3.3V |
| **4.75V-5.25V Main power input** | Main power input | **VBUS** | 83 | 84 | **VCC_3V3_S3** | POWER VCC_3V Out | **3.3V +/-2.5% Power Output max 300mA per pin for a total of 600mA.** |
| **4.75V-5.25V Main power input** | Main power input | **VBUS** | 85 | 86 | **VCC_3V3_S3** | POWER VCC_3V Out | **3.3V +/-2.5% Power Output max 300mA per pin for a total of 600mA.** |
| **4.75V-5.25V Main power input** | Main power input | **VBUS** | 87 | 88 | **VCC_1V8_S3** | POWER VCC_3V Out | **1.8V +/-2.5% Power Output max 300mA per pin for a total of 600mA.** |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Floatimg | | Reserved | 89 | 90 | **VCC_1V8_S3** | POWER VCC_3V Out | **1.8V +/-2.5% Power Output max 300mA per pin for a total of 600mA.** |
| Floatimg | | Reserved | 91 | 92 | PMIC_RESET_L | RESET,PMU IN And RK3588S | Bidirectional pin. Can be driven low to Reset the CM5 CPU. |
| A low on this pin forces booting , Internally pulled up via IO to1.8V | SARADC_IN0_BOOT | SARADC_IN0 | 93 | 94 | SARADC_IN3 | SARADC_IN3 | 12-bit ADC in signal,MAX 1.8V |
| GPIO4_B6, for 3.3V | BT1120_D12/SATA0_ACT_LED_M0/DDRPHY_CH3_DTB2/I2C5_SCL_M1/PWM13_M1/SPI3_MOSI_M1/GPIO4_B6_d | GPIO4_B6 | 95 | 96 | SARADC_IN2 | SARADC_IN2 | 12-bit ADC in signal,MAX 1.8V |
| GPIO4_A7, for 3.3V | CIF_D7/BT1120_D7/I2S1_SDI2_M0/DDRPHY_CH1_DTB3/I2C5_SDA_M2/SPI2_CS0_M1/GPIO4_A7_d | GPIO4_A7 | 97 | 98 | **GND** | | **Ground (0V)** |
| Input. Drive low to power off or power on ,CM5. Internally pulled up with a 45K to VBUS | POWER KEY( PWRON_L To GND). | PWRON_L | 99 | 100 | GPIO4_A5 | CIF_D5/BT1120_D5/I2S1_SDI0_M0/DDRPHY_CH1_DTB_1/I2C3_SDA_M2/UART3_TX_M2/SPI2_MOSI_M1/GPIO4_A5_d | GPIO4_A5, for 3.3V |

## OPi_CM5_CORE JP2 100PIN

| **Description** | | **Signal** | **PIN NO.** | | **Signal** | | **Description** |
|---|---|---|---|---|---|---|---|
| TYPEC0_USB20_OTG_ID | TYPEC0_USB20_OTG_ID | TYPEC0_USB20_OTG_ID | 1 | 2 | GPIO1_A0 | PCIE20X1_1_CLKREQN_M2/DP0_HPDIN_M2/I2C2_SDA_M4/UART6_RX_M1/SPI4_MISO_M2/GPIO1_A0_d | PCIE_CLKREQN GPIO1_A0 for 3.3V |
| TYPEC0_USB20_OTG_DM | TYPEC0_USB20_OTG_DM | TYPEC0_USB20_OTG_DM | 3 | 4 | GPIO1_A1 | PCIE20X1_1_WAKEN_M2/I2C2_SCL_M4/UART6_TX_M1/SPI4_MOSI_M2/GPIO1_A1_d | PCIE_WAKEN GPIO1_A1 for 3.3V |
| TYPEC0_USB20_OTG_DP | TYPEC0_USB20_OTG_DP | TYPEC0_USB20_OTG_DP | 5 | 6 | GPIO4_A6 | CIF_D6/BT1120_D6/I2S1_SDI1_M0/DDRPHY_CH1_DTB2/I2C5_SCL_M2/UART3_RX_M2/SPI2_CLK_M1/GPIO4_A6_d | GPIO4_A6 for 3.3V |
| **Ground (0V)** | | **GND** | **7** | **8** | **GND** | | **Ground (0V)** |
| PCIE20_PERSTn_3V3_L GPIO4_A2 for 3.3V | CIF_D2/BT1120_D2/I2S1_LRCK_M0/PCIE20X1_1_PERSTN_M1/DDRPHY_CH0_DTB2/SPI0_CLK_M1/GPIO4_A2_d | GPIO4_A2 | 9 | 10 | PCIE20_2_REFCLKP | PCIE20_2_REFCLKP | PCIE20_2_REFCLKP, |
| Floatimg | | Reserved | 11 | 12 | PCIE20_2_REFCLKN | PCIE20_2_REFCLKN | PCIE20_2_REFCLKN |
| **Ground (0V)** | | **GND** | **13** | **14** | **GND** | | **Ground (0V)** |
| MIPI_CSI0_RX_D0N | MIPI_CSI0_D0N | MIPI_CSI0_D0N | 15 | 16 | PCIE20_2_RXP/SATA30_2_RXP | PCIE20_2_RXP/SATA30_2_RXP/USB30_2_SSRXP | PCIE20_2_RXP SATA30_2_RXP |
| MIPI_CSI0_RX_D0P | MIPI_CSI0_D0P | MIPI_CSI0_D0P | 17 | 18 | PCIE20_2_RXN/SATA30_2_RXN | PCIE20_2_RXN/SATA30_2_RXN/USB30_2_SSRXN | PCIE20_2_RXN SATA30_2_RXN |
| **Ground (0V)** | | **GND** | **19** | **20** | **GND** | | **Ground (0V)** |
| MIPI_CSI0_RX_D1N | MIPI_CSI0_D1N | MIPI_CSI0_D1N | 21 | 22 | PCIE20_2_TXP/SATA30_2_TXP | PCIE20_2_TXP/SATA30_2_TXP/USB30_2_SSTXP | PCIE20_2_TXP SATA30_2_TXP |
| MIPI_CSI0_RX_D1P | MIPI_CSI0_D1P | MIPI_CSI0_D1P | 23 | 24 | PCIE20_2_TXN/SATA30_2_TXN | PCIE20_2_TXN/SATA30_2_TXN/USB30_2_SSTXN | PCIE20_2_TXN SATA30_2_TXN |
| Ground (0V) | | GND | 25 | 26 | GND | | Ground (0V) |
| MIPI_CSI0_RX_CLK0N | MIPI_CSI0_CLK0N | MIPI_CSI0_CLK0N | 27 | 28 | MIPI_DPHY0_RX_D0N | MIPI_DPHY0_RX_D0N/MIPI_CPHY0_RX_TRIO0_A | MIPI_DPHY0_RX_D0N |
| MIPI_CSI0_RX_CLK0P | MIPI_CSI0_CLK0P | MIPI_CSI0_CLK0P | 29 | 30 | MIPI_DPHY0_RX_D0P | MIPI_DPHY0_RX_D0P/MIPI_CPHY0_RX_TRIO0_B | MIPI_DPHY0_RX_D0P |
| **Ground (0V)** | | **GND** | **31** | **32** | **GND** | | **Ground (0V)** |
| MIPI_CSI0_RX_D2N | MIPI_CSI0_D2N | MIPI_CSI0_D2N | 33 | 34 | MIPI_DPHY0_RX_D1N | MIPI_DPHY0_RX_D1N/MIPI_CPHY0_RX_TRIO0_C | MIPI_DPHY0_RX_D1N |
| MIPI_CSI0_RX_D2N | MIPI_CSI0_D2N | MIPI_CSI0_D2N | 35 | 36 | MIPI_DPHY0_RX_D1P | MIPI_DPHY0_RX_D1P/MIPI_CPHY0_RX_TRIO1_A | MIPI_DPHY0_RX_D1P |
| **Ground (0V)** | | **GND** | **37** | **38** | **GND** | | **Ground (0V)** |
| MIPI_CSI0_RX_D3N | MIPI_CSI0_D3N | MIPI_CSI0_D3N | 39 | 40 | MIPI_DPHY0_RX_CLKN | MIPI_DPHY0_RX_CLKN/MIPI_CPHY0_RX_TRIO1_B | MIPI_DPHY0_RX_CLKN |
| MIPI_CSI0_RX_D3P | MIPI_CSI0_D3P | MIPI_CSI0_D3P | 41 | 42 | MIPI_DPHY0_RX_CLKP | MIPI_DPHY0_RX_CLKP/MIPI_CPHY0_RX_TRIO1_C | MIPI_DPHY0_RX_CLKP |
| Floatimg | | Reserved | 43 | 44 | **GND** | | **Ground (0V)** |
| MIPI_CSI0_RX_CLK1N | MIPI_CSI0_CLK1N | MIPI_CSI0_CLK1N | 45 | 46 | Reserved | | Floatimg |
| MIPI_CSI0_RX_CLK1P | MIPI_CSI0_CLK1P | MIPI_CSI0_CLK1P | 47 | 48 | Reserved | | Floatimg |
| HDMI0_TX_ON_H GPIO4_B5 for 3.3V | BT1120_D11/DDRPHY_CH3_DTB1/UART9_RX_M1/PWM12_M1/SPI3_MISO_M1/GPIO4_B5_d | GPIO4_B5 | 49 | 50 | **GND** | | **Ground (0V)** |
| HDMI0_TX_CEC_M0 GPIO4_C1 for 3.3V | BT1120_D15/SPDIF1_TX_M2/PCIE20X1_2_PERSTN_M1/HDMI_TX0_CEC_M0/I2C8_SDA_M3/PWM6_M1/SPI3_CS1_M1/GPIO4_C1_d | HDMI_TX0_CEC_M0 | 51 | 52 | Reserved | | Floatimg |
| HDMI0_SBDN/HPD for HDMI V2.1 HDMI0_HPD for HDMI V2.0 | HDMI_TX0_SBDN/EDP_TX0_AUXN | HDMI_TX0_SBDN | 53 | 54 | Reserved | | Floatimg |
| **Ground (0V)** | | **GND** | 55 | 56 | **GND** | | **Ground (0V)** |
| MIPI_DPHY0_TX_D0N | MIPI_DPHY0_TX_D0N/MIPI_CPHY0_TX_TRIO0_A | MIPI_DPHY0_TX_D0N | 57 | 58 | HDMI0_SBDP | | HDMI0_SBDP,for HDMI V2.1 |
| MIPI_DPHY0_TX_D0P | MIPI_DPHY0_TX_D0P/MIPI_CPHY0_TX_TRIO0_B | MIPI_DPHY0_TX_D0P | 59 | 60 | Reserved | | Floatimg |
| **Ground (0V)** | | **GND** | **61** | **62** | **GND** | | **Ground (0V)** |
| MIPI_DPHY0_TX_D1N | MIPI_DPHY0_TX_D1N/MIPI_CPHY0_TX_TRIO0_C | MIPI_DPHY0_TX_D1N | 63 | 64 | Reserved | | Floatimg |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| MIPI_DPHY0_TX_D1P | MIPI_DPHY0_TX_D1P/MIPI_CPHY0_TX_TRIO1_A | MIPI_DPHY0_TX_D1P | 65 | 66 | Reserved | | Floatimg |
| **Ground (0V)** | | **GND** | **67** | **68** | **GND** | | **Ground (0V)** |
| MIPI_DPHY0_TX_CLKN | MIPI_DPHY0_TX_CLKN/MIPI_CPHY0_TX_TRIO1_B | MIPI_DPHY0_TX_CLKN | 69 | 70 | HDMI_TX0_D2P | HDMI_TX0_D2P/EDP_TX0_D2P | HDMI0_TX2P |
| MIPI_DPHY0_TX_CLKP | MIPI_DPHY0_TX_CLKP/MIPI_CPHY0_TX_TRIO1_C | MIPI_DPHY0_TX_CLKP | 71 | 72 | HDMI_TX0_D2N | HDMI_TX0_D2N/EDP_TX0_D2N | HDMI0_TX2N |
| **Ground (0V)** | | **GND** | **73** | **74** | **GND** | | **Ground (0V)** |
| MIPI_DPHY1_TX_D0N | MIPI_DPHY1_TX_D0N/MIPI_CPHY1_TX_TRIO0_A | MIPI_DPHY1_TX_D0N | 75 | 76 | HDMI_TX0_D1P | HDMI_TX0_D1P/EDP_TX0_D1P | HDMI0_TX1P |
| MIPI_DPHY1_TX_D0P | MIPI_DPHY1_TX_D0P/MIPI_CPHY1_TX_TRIO0_B | MIPI_DPHY1_TX_D0P | 77 | 78 | HDMI_TX0_D1N | HDMI_TX0_D1N/EDP_TX0_D1N | HDMI0_TX1N |
| **Ground (0V)** | | **GND** | **79** | **80** | **GND** | | **Ground (0V)** |
| MIPI_DPHY1_TX_D1N | MIPI_DPHY1_TX_D1N/MIPI_CPHY1_TX_TRIO0_C | MIPI_DPHY1_TX_D1N | 81 | 82 | HDMI_TX0_D0P | HDMI_TX0_D0P/EDP_TX0_D0P | HDMI0_TX0P |
| MIPI_DPHY1_TX_D1P | MIPI_DPHY1_TX_D1P/MIPI_CPHY1_TX_TRIO1_A | MIPI_DPHY1_TX_D1P | 83 | 84 | HDMI_TX0_D0N | HDMI_TX0_D0N/EDP_TX0_D0N | HDMI0_TX0N |
| **Ground (0V)** | | **GND** | **85** | **86** | **GND** | | **Ground (0V)** |
| MIPI_DPHY1_TX_CLKN | MIPI_DPHY1_TX_CLKN/MIPI_CPHY1_TX_TRIO1_B | MIPI_DPHY1_TX_CLKN | 87 | 88 | HDMI_TX0_D3P | HDMI_TX0_D3P/EDP_TX0_D3P | HDMI0_TX3P |
| MIPI_DPHY1_TX_CLKP | MIPI_DPHY1_TX_CLKP/MIPI_CPHY1_TX_TRIO1_C | MIPI_DPHY1_TX_CLKP | 89 | 90 | HDMI_TX0_D3N | HDMI_TX0_D3N/EDP_TX0_D3N | HDMI0_TX3N |
| **Ground (0V)** | | **GND** | **91** | **92** | **GND** | | **Ground (0V)** |
| MIPI_DPHY1_TX_D2N | MIPI_DPHY1_TX_D2N/MIPI_CPHY1_TX_TRIO2_A | MIPI_DPHY1_TX_D2N | 93 | 94 | MIPI_DPHY1_TX_D3N | MIPI_DPHY1_TX_D3N/MIPI_CPHY1_TX_TRIO2_C | MIPI_DPHY1_TX_D3N |
| MIPI_DPHY1_TX_D2P | MIPI_DPHY1_TX_D2P/MIPI_CPHY1_TX_TRIO2_B | MIPI_DPHY1_TX_D2P | 95 | 96 | MIPI_DPHY1_TX_D3P | MIPI_DPHY1_TX_D3P/NO_USE | MIPI_DPHY1_TX_D3P |
| **Ground (0V)** | | **GND** | **97** | **98** | **GND** | | **Ground (0V)** |
| HDMI_TX0_SDA_M0<br>GPIO4_C0, for 3.3V | BT1120_D14/PCIE20X1_2_WAKEN_M1/HDMI_TX0_SDA_M0/I2C8_SCL_M3/SPI3_CS0_M1/GPIO4_C0_u | HDMI_TX0_SDA_M0 | 99 | 100 | HDMI_TX0_SCL_M0 | BT1120_D13/PCIE20X1_2_CLKREQN_M1/HDMI_TX0_SCL_M0/DDRPHY_CH3_DTB3/I2C5_SDA_M1/SPI3_CLK_M1/GPIO4_B7_u | HDMI_TX0_SCL_M0<br>GPIO4_B7, for 3.3V |

## OPi_CM5_CORE JP3 100PIN

| **Description** | | **Signal** | **PIN NO.** | | **Signal** | | **Description** |
|---|---|---|---|---|---|---|---|
| USB20_HOST1_DP | USB20_HOST1_DP | USB20_HOST1_DP | 1 | 2 | TYPEC0_SSTX2P | TYPEC0_SSTX2P/DP0_TX3P | TYPEC0_SSTX2P |
| USB20_HOST1_DM | USB20_HOST1_DM | USB20_HOST1_DM | 3 | 4 | TYPEC0_SSTX2N | TYPEC0_SSTX2N/DP0_TX3N | TYPEC0_SSTX2N |
| **Ground (0V)** | | **GND** | **5** | **6** | **GND** | | **Ground (0V)** |
| TYPEC_SBU1 | TYPEC0_SBU1/DP0_AUXP | TYPEC0_SBU1 | 7 | 8 | TYPEC0_SSRX2P | TYPEC0_SSRX2P/DP0_TX2P | TYPEC0_SSRX2P |
| TYPEC_SBU2 | TYPEC0_SBU2/DP0_AUXN | TYPEC0_SBU2 | 9 | 10 | TYPEC0_SSRX2N | TYPEC0_SSRX2N/DP0_TX2N | TYPEC0_SSRX2N |
| **Ground (0V)** | | **GND** | **11** | **12** | **GND** | | **Ground (0V)** |
| USB20_HOST0_DP | USB20_HOST0_DP | USB20_HOST0_DP | 13 | 14 | TYPEC0_SSTX1N | TYPEC0_SSTX1N/DP0_TX1N | TYPEC0_SSTX1N |
| USB20_HOST0_DM | USB20_HOST0_DM | USB20_HOST0_DM | 15 | 16 | TYPEC0_SSTX1P | TYPEC0_SSTX1P/DP0_TX1P | TYPEC0_SSTX1P |
| **Ground (0V)** | | **GND** | **17** | **18** | **GND** | | **Ground (0V)** |
| TYPEC0_SSRX1P | TYPEC0_SSRX1P/DP0_TX0P | TYPEC0_SSRX1P | 19 | 20 | GPIO3_C6 | CIF_D10/SPI3_MISO_M3/GPIO3_C6_u | GPIO3_C6 for1.8V |
| TYPEC0_SSRX1P | TYPEC0_SSRX1P/DP0_TX0P | TYPEC0_SSRX1P | 21 | 22 | GPIO3_C7 | CIF_D11/PCIE20X1_2_CLKREQN_M0/HDMI_TX0_SCL_M2/I2C5_SCL_M0/SPI3_MOSI_M3/GPIO3_C7_u | PCIE20x1_2_CLKREQn_M0<br>GPIO3_C7, for 1.8V |
| **Ground (0V)** | | **GND** | 23 | 24 | GPIO3_D1 | CIF_D13/PCIE20X1_2_PERSTN_M0/UART4_TX_M1/PWM9_M2/SPI0_MISO_M3/GPIO3_D1_d | PCIE20x1_2_PERSTn_M0<br>GPIO3_D1, for 1.8V |
| MIPI_DPHY1_RX_CLKP | MIPI_DPHY1_RX_CLKP/MIPI_CPHY1_RX_TRIO1_C | MIPI_DPHY1_RX_CLKP | 25 | 26 | GPIO3_D5 | MCU_JTAG_TMS_M1/UART9_TX_M2/PWM11_IR_M3/SPI0_CS1_M3/GPIO3_D5_d | UART9_TX_M2_BT<br>GPIO3_D5, for 1.8V |
| MIPI_DPHY1_RX_CLKN | MIPI_DPHY1_RX_CLKN/MIPI_CPHY1_RX_TRIO1_B | MIPI_DPHY1_RX_CLKN | 27 | 28 | GPIO3_D2 | CIF_D14/I2C7_SCL_M2/UART9_RTSN_M2/SPI0_MOSI_M3/GPIO3_D2_d | UART9_RTSN_M2_BT<br>GPIO3_D2, for 1.8V |
| **Ground (0V)** | | **GND** | 29 | 30 | GPIO3_D4 | HDMI_TX0_HPD_M1/MCU_JTAG_TCK_M1/UART9_RX_M2/SPI0_CS0_M3/GPIO3_D4_d | UART9_RX_M2_BT<br>GPIO3_D4, for 1.8V |
| MIPI_DPHY1_RX_D1P | MIPI_DPHY1_RX_D1P/MIPI_CPHY1_RX_TRIO1_A | MIPI_DPHY1_RX_D1P | 31 | 32 | **GND** | | **Ground (0V)** |
| MIPI_DPHY1_RX_D1N | MIPI_DPHY1_RX_D1N/MIPI_CPHY1_RX_TRIO0_C | MIPI_DPHY1_RX_D1N | 33 | 34 | GPIO3_B7 | GMAC1_PTP_REF_CLK/I2C3_SCL_M1/SPI1_MOSI_M1/GPIO3_B7_d | GPIO3_B7,for 1.8V |
| **Ground (0V)** | | **GND** | 35 | 36 | GPIO3_C5 | CIF_D9/FSPI_CS1N_M2/CAN2_TX_M0/UART5_RX_M1/SPI3_CS1_M3/GPIO3_C5_u | GPIO3_C5,for 1.8V |
| MIPI_DPHY1_RX_D0P | MIPI_DPHY1_RX_D0P/MIPI_CPHY1_RX_TRIO0_B | MIPI_DPHY1_RX_D0P | 37 | 38 | GPIO3_C4 | CIF_D8/FSPI_CS0N_M2/CAN2_RX_M0/UART5_TX_M1/SPI3_CS0_M3/GPIO3_C4_u | GPIO3_C4,for 1.8V |
| MIPI_DPHY1_RX_D0N | MIPI_DPHY1_RX_D0N/MIPI_CPHY1_RX_TRIO0_A | MIPI_DPHY1_RX_D0N | 39 | 40 | GPIO3_D3 | CIF_D15/I2C7_SDA_M2/UART9_CTSN_M2/PWM10_M2/SPI0_CLK_M3/GPIO3_D3_d | UART9_CTSN_M2_BT<br>GPIO3_D3, for 1.8V |
| **Ground (0V)** | | **GND** | 41 | 42 | GPIO3_A1 | GMAC1_TXD3/SDIO_D1_M1/I2S3_SCLK/AUDDSM_LN/FSPI_D2_M2/I2C6_SCL_M4/PWM11_IR_M0/SPI4_MOSI_M1/GPIO3_A1_u | GMAC_TXD3/SDIO_D1<br>GPIO3_A1, for 1.8V |
| PCIE20x1_2_WAKEn_M0<br>GPIO3_D0, for 1.8V | CIF_D12/PCIE20X1_2_WAKEN_M0/HDMI_TX0_SDA_M2/I2C5_SDA_M0/UART4_RX_M1/PWM8_M2/SPI3_CLK_M3/GPIO3_D0_u | GPIO3_D0 | 43 | 44 | **GND** | | **Ground (0V)** |
| **Ground (0V)** | | **GND** | 45 | 46 | GPIO3_B3 | GMAC1_TXD0/I2S2_SDO_M1/UART2_RTSN/GPIO3_B3_u | GMAC_TXD0<br>GPIO3_B3, for 1.8V |
| GMAC_TXD2/SDIO_D0<br>GPIO3_A0, for 1.8V | GMAC1_TXD2/SDIO_D0_M1/I2S3_MCLK/FSPI_D0_M2/I2C6_SDA_M4/PWM10_M0/SPI4_MISO_M1/GPIO3_A0_u | GPIO3_A0 | 47 | 48 | GPIO3_B2 | GMAC1_TXER/I2S2_SDI_M1/UART2_RX_M2/PWM3_IR_M1/GPIO3_B2_d | GMAC_TXER/PWM3_IR<br>GPIO3_B2, for 1.8V |
| GMAC_TXD1<br>GPIO3_B4, for 1.8V | GMAC1_TXD1/I2S2_MCLK_M1/UART2_CTSN/GPIO3_B4_u | GPIO3_B4 | 49 | 50 | GPIO3_C1 | GMAC1_PPSCLK/UART7_RX_M1/SPI1_CLK_M1/GPIO3_C1_d | GPIO3_C1,for 1.8V |
| GMAC_TXEN/PWM12<br>GPIO3_B5, for 1.8V | GMAC1_TXEN/I2S2_SCLK_M1/CAN1_RX_M0/UART3_TX_M1/PWM12_M0/GPIO3_B5_u | GPIO3_B5 | 51 | 52 | GPIO3_A6 | ETH1_REFCLKO_25M/MIPI_CAMERA1_CLK_M1/I2C4_SCL_M0/GPIO3_A6_d | MIPI_CLK1<br>GPIO3_A6, for 1.8V |
| GMAC_MDIO<br>GPIO3_C3, for 1.8V | GMAC1_MDIO/MIPI_TE1/I2C8_SDA_M4/UART7_CTSN_M1/PWM15_IR_M0/SPI1_CS1_M1/GPIO3_C3_d | GPIO3_C3 | 53 | 54 | **GND** | | **Ground (0V)** |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **Ground (0V)** | | **GND** | 55 | 56 | GPIO3_B0 | GMAC1_RXD1/MIPI_CAMERA3_CLK_M1/PWM9_M0/GPIO3_B0_u | GMAC_RXD1/MIPI_CLK3/PWM9<br>GPIO3_B0, for 1.8V |
| GMAC_MCLK/PWM13<br>GPIO3_B6, for 1.8V | GMAC1_MCLKINOUT/I2S2_LRCK_M1/CAN1_TX_M0/UART3_RX_M1/PWM13_M0/GPIO3_B6_d | GPIO3_B6 | 57 | 58 | GPIO3_A5 | GMAC1_RXCLK/SDIO_CLK_M1/MIPI_CAMERA0_CLK_M1/FSPI_CLK_M2/I2C4_SDA_M0/UART8_CTSN_M1/GPIO3_A5_d | GMAC_RXCLK/SDIO_CLK<br>GPIO3_A5, for 1.8V |
| GMAC_RXDV/MIPI_CLK4/PWM2<br>GPIO3_B1, for 1.8V | GMAC1_RXDV_CRS/MIPI_CAMERA4_CLK_M1/UART2_TX_M2/PWM2_M1/GPIO3_B1_d | GPIO3_B1 | 59 | 60 | GPIO3_A7 | GMAC1_RXD0/MIPI_CAMERA2_CLK_M1/PWM8_M0/GPIO3_A7_u | GMAC_RXD0/MIPI_CLK2/PWM8<br>GPIO3_A7, for 1.8V |
| GMAC_RXD2/SDIO_D2<br>GPIO3_A2, for 1.8V | GMAC1_RXD2/SDIO_D2_M1/I2S3_LRCK/AUDDSM_LP/FSPI_D2_M2/UART8_TX_M1/SPI4_CLK_M1/GPIO3_A2_u | GPIO3_A2 | 61 | 62 | GPIO3_A4 | GMAC1_TXCLK/SDIO_CMD_M1/I2S3_SDI/AUDDSM_RP/UART8_RTSN_M1/SPI4_CS1_M1/GPIO3_A4_d | GMAC_TXCLK/SDIO_CMD<br>GPIO3_A4, for 1.8V |
| GMAC_RXD3/SDIO_D3<br>GPIO3_A3, for 1.8V | GMAC1_RXD3/SDIO_D3_M1/I2S3_SDO/AUDDSM_RN/FSPI_D3_M2/UART8_RX_M1/SPI4_CS0_M1/GPIO3_A3_u | GPIO3_A3 | 63 | 64 | **GND** | | **Ground (0V)** |
| **Ground (0V)** | | **GND** | 65 | 66 | GPIO0_B0 | SPI2_CS1_M2/I2C1_SCL_M1/UART0_RX_M1/GPIO0_B0_z | GPIO0_B0, for 1.8V |
| REFCLK_OUT/GPIO0_A0 , for 1.8V | REFCLK_OUT/GPIO0_A0_d | GPIO0_A0 | 67 | 68 | I2S0_MCLK | I2S0_MCLK/I2C6_SDA_M1/UART3_RTSN/PWM3_IR_M2/SPI4_CLK_M0/GPIO1_C2_d | I2S0_MCLK<br>GPIO1_C2, for 1.8V |
| I2S0_SRLK<br>GPIO1_C2, for 1.8V | I2S0_MCLK/I2C6_SDA_M1/UART3_RTSN/PWM3_IR_M2/SPI4_CLK_M0/GPIO1_C2_d | GPIO1_C2 | 69 | 70 | I2S0_SDO0 | I2S0_SDO0/I2C4_SCL_M4/UART4_CTSN/GPIO1_C7_d | I2S0_SDO0<br>GPIO1_C7, for 1.8V |
| I2S0_SCLK<br>GPIO1_C5, for 1.8V | I2S0_LRCK/I2C2_SCL_M3/UART4_RTSN/GPIO1_C5_d | GPIO1_C5 | 71 | 72 | GPIO3_C0 | GMAC1_PPSTRIG/I2C3_SDA_M1/UART7_TX_M1/SPI1_MISO_M1/GPIO3_C0_d | GPIO3_C0, for 1.8V |
| I2S0_SDI0<br>GPIO1_D4, for 1.8V | I2S0_SDI0/GPIO1_D4_d | GPIO1_D4 | 73 | 74 | **GND** | | **Ground (0V)** |
| **Ground (0V)** | | **GND** | 75 | 76 | PCIE20_0_REFCLKP | PCIE20_0_REFCLKP | PCIE20_0_REFCLKP |
| GPIO0_D5, for 1.8V | I2S1_SDO3_M1/CPU_BIG1_AVS/I2C1_SDA_M2/CAN2_TX_M1/HDMI_TX0_SCL_M1/SPI3_CS1_M2/SATA_MP_SWITCH/GPIO0_D5_u | GPIO0_D5 | 77 | 78 | PCIE20_0_REFCLKN | PCIE20_0_REFCLKN | PCIE20_0_REFCLKN |
| GPIO0_C7, for 1.8V | I2S1_SDI2_M1/PDM0_SDI0_M1/I2C6_SDA_M0/UART1_RTSN_M2/PWM6_M0/SPI0_MISO_M0/GPIO0_C7_d | GPIO0_C7 | 79 | 80 | **GND** | | **Ground (0V)** |
| GPIO0_D4, for 1.8V | I2S1_SDO2_M1/PDM0_SDI2_M1/PWM3_IR_M0/I2C1_SCL_M2/CAN2_RX_M1/HDMI_TX0_SDA_M1/SPI3_CS0_M2/SATA_CPDET/GPIO0_D4_u | GPIO0_D4 | 81 | 82 | PCIE20_0_TXP | PCIE20_0_TXP/SATA30_0_TXP | PCIE20_0_TXP |
| GPIO0_C4, for 1.8V | PDM0_CLK1_M1/PWM2_M0/UART0_RX_M0/I2C4_SDA_M2/DP0_HPDIN_M1/GPIO0_C4_d | GPIO0_C4 | 83 | 84 | PCIE20_0_TXN | PCIE20_0_TXN/SATA30_0_TXN | PCIE20_0_TXN |
| GPIO0_D3, for 1.8V | LITCPU_AVS/SPI3_CLK_M2/GPIO0_D3_u | GPIO0_D3 | 85 | 86 | **GND** | | **Ground (0V)** |
| **Ground (0V)** | | **GND** | 87 | 88 | PCIE20_0_RXN | PCIE20_0_RXN/SATA30_0_RXN | PCIE20_0_RXN |
| GPIO0_D0, for 1.8V | I2S1_SDI3_M1/PDM0_SDI1_M1/I2C6_SCL_M0/UART1_CTSN_M2/PWM7_IR_M0/SPI3_MISO_M2/GPIO0_D0_d | GPIO0_D0 | 89 | 90 | PCIE20_0_RXP | PCIE20_0_RXP/SATA30_0_RXP | PCIE20_0_RXP |
| GMAC_MDC<br>GPIO3_C2, for 1.8V | GMAC1_MDC/MIPI_TE0/I2C8_SCL_M4/UART7_RTSN_M1/PWM14_M0/SPI1_CS0_M1/GPIO3_C2_d | GPIO3_C2 | 91 | 92 | **GND** | | **Ground (0V)** |
| GPIO4_A3, for 3.3V | CIF_D3/BT1120_D3/I2S1_SCLK_M0/DDRPHY_CH0_DTB_3/UART0_TX_M2/GPIO4_A3_d | GPIO4_A3 | 93 | 94 | GPIO0_C5 | I2S1_SDI0_M1/GPU_AVS/UART0_TX_M0/I2C4_SCL_M2/PWM4_M0/GPIO0_C5_u | GPIO0_C5, for 1.8V |
| CAN1_TX_M1/GPIO4_B3, for 3.3V | CIF_VSYNC/BT1120_D9/I2S1_SDO2_M0/PCIE20X1_2_BUTTON_RSTN/DDRPHY_CH2_DTB3/I2C7_SDA_M3/UART8_CTSN_M0/PWM15_IR_M1/CAN1_TX_M1/GPIO4_B3_u | GPIO4_B3 | 95 | 96 | GPIO0_B2 | CLK32K_IN/CLK32K_OUT0/GPIO0_B2_u | CLK32K_IN/GPIO0_B2, for 1.8V |
| CAN1_RX_M1/GPIO4_B2, for 3.3V | CIF_HREF/BT1120_D8/I2S1_SDO1_M0/PCIE20X1_1_BUTTON_RSTN/DDRPHY_CH2_DTB2/I2C7_SCL_M3/UART8_RTSN_M0/PWM14_M1/SPI0_CS0_M1/CAN1_RX_M1/GPIO4_B2_u | GPIO4_B2 | 97 | 98 | GPIO0_C6 | I2S1_SDI1_M1/NPU_AVS/UART0_RTSN/PWM5_M1/SPI0_CLK_M0/SATA_CP_POD/GPIO0_C6_u | GPIO0_C6, for 1.8V |
| **Ground (0V)** | | **GND** | **99** | **100** | **GND** | | **Ground (0V)** |